

Connecting Real Sensors to B.R.A.V.E.

Your exact 5-sensor setup — what to change to make it live

Your sensor map

Vibration MPU6050	vibration_rms	Vibration card
Temperature DS18B20	temperature_c	Temperature card
Roller Speed Hall Sensor	rpm	Roller Speed card
Belt Position HC-SR04	position_mm	Belt Position card
Connection status ESP32	(automatic)	"ESP32 CONNECTED" pill — no field to send, see Step 2

Everything else the dashboard shows (joints, anomaly scores, fault location, predictive maintenance, alerts) is still simulated and unrelated to this hardware change.

Already done for you — `dashboard_gateway.py` is sitting in the project root, next to `app.py`. You don't need to create or edit any files; just follow the steps below.

Step 1 — Set the serial connection

Open `config.py` and set these two lines to match your hardware (everything else in the file can stay as-is):

```
SERIAL_PORT = "COM5"
BAUD_RATE = 115200
```

Find the COM port in Windows Device Manager → Ports (COM & LPT) while the ESP32 is plugged in.

Step 2 — Make the ESP32 send this JSON

One line per reading, over serial — just the 4 sensor fields plus a health flag per sensor. This is the only firmware-side change needed:

```
{
  "vibration_rms": 0.74,
  "temperature_c": 41.2,
  "rpm": 81.5,
  "position_mm": 640,
  "mpu6050_ok": true,
  "temp_ok": true
}
```

Any line that doesn't start with `{` is treated as debug text and ignored, so `Serial.println()` calls for debugging are safe to leave in. "Connection status" needs no field of its own — `dashboard_gateway.py` stamps every reading it receives with a timestamp, and the dashboard shows "ESP32 DISCONNECTED" automatically if that timestamp goes stale for more than 5 seconds.

Step 3 — Install pyserial and run the gateway

```
pip install pyserial
```

Close the Arduino Serial Monitor first — only one program can hold the COM port at a time. Then, in a terminal separate from the one running Streamlit:

```
python dashboard_gateway.py
```

Leave it running. `dashboard_state.json` should appear in the project folder within a few seconds, and the gateway prints one line per reading.

Step 4 — Switch the dashboard to live mode

In `data/live_data.py`, change:

```
DEMO_MODE = False
```

Refresh the dashboard. Control Room and Live Sensors now move with your real hardware — the other 5 screens keep using demo data until wired separately.